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27 October 1953
SUPERSEDING
Army 96-131-231
10 September 1945

MILITARY SPECIFICATION

GUN, FOR PORTABLE FLAME THROWER, M2A1

1. SCOPE

1.1 This specification covers one type of gun for use in the controlled firing and igniting of flame throwing fuel.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111 - Tape; Paper, Gummed (Kraft).
VV-D-361 - Diesel Fuel.

MILITARY

JAN-P-106 - Boxes; Wood, Nailed (Military Type).
JAN-P-108 - Boxes, Fiberboard Corrugated and Solid (Military Type).
MIL-L-644 - Lubricating-Oil, Preservative, Special.
MIL-P-10577 - Packaging of Components and Subassemblies, Domestic Shipment to Assembly Plants.

STANDARDS

MILITARY

MIL-STD-10 - Surface Roughness, Waviness, and Lay.
MIL-STD-129 - Marking of Shipments.

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DRAWINGS

CHEMICAL CORPS

C81-1-403 - Gun, Portable Flame Thrower, M2A1 - Bill of Material - Sheet 1 of 2 Sheets.
C81-1404 - Gun, Portable Flame Thrower, M2A1 - Bill of Material - Sheet 2 of 2 Sheets.
B81-2-20 - Box, Portable Flame Thrower Gun Packing - Packing Diagram and Bill of Material.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procurin agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - All materials and components used in the manufacture and assembly of the gun shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture and assembly. - The gun shall be manufactured and assembled as specified on Drawing D81-1405 and subsidiary drawings listed on Drawings C81-1-403 and C81-1-404. Surface roughness, waviness, and lay specified by symbols on detail drawings shall comply with the applicable classes specified in Standard MIL-STD-10.

3.3 Valve assembly. - The complete valve shall be assembled as specified on Drawing D81-1-407 which is listed on Drawing C81-1-403. All bearing surfaces of the pivots, springs, and locking surfaces of the valve handle shall be lubricated at assembly with special preservative lubricating oil conforming to Specification MIL-L-644.

3.3.1 Valve assembly adjustment. - The compression in the valve spring, Drawing E81-1-435 which is listed on Drawing C81-1-403, shall be adjusted to mee the requirements specified in 3.3.1.1 and 3.2.1.2 by turning the needle valve, Drawing B81-1-411 which is listed on Drawing C81-1-404, in the yoke block, Drawing A81-1-431 which is listed on Drawing C81-1-403, and testing as specified in 4.4.1. After adjustment, the locknut on the needle valve shall be jammed on the yoke block to prevent further turning of the needle valve in the yoke block.

3.3.1.1 Needle travel. - The needle travel shall be tested for location when th operating lever is pulled back for its full stroke as specified in 4.4.1.1.

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3.3.1.2 Operating lever. - The force required to depress the operating lever for its full stroke shall not exceed 40 pounds when the force is applied at the center of the lowest finger notch (farthest from pivot) when tested as specified in 4.4.1.2. A 15-pound force so applied shall not open the valve.

3.3.2 Safety. - The operating lever, Drawing B81-1-426 which is listed on Drawing C81-1-403, shall not move by applying hand pressure, unless the grip safety, Drawing B81-1-425 which is listed on Drawing C81-1-403, is first depressed when tested as specified in 4.4.1.3.

3.4 Ignition head assembly. - The ignition head assembly shall be assembled as specified on Drawing D81-1-406 which is listed on Drawing C81-1-403 and tested as specified in 4.4.2. The ignition cylinder shall not be assembled in the head at the time of manufacture.

3.4.1 Spring case. - The spring case shall be assembled as specified on Drawing B81-1-444 which is listed on Drawing C81-1-403. With one case fixed, the other case shall be capable of being rotated not less than 2-1/4 revolutions under spring tension. The torque required to wind the spring shall be not less than 2-1/2 inch-pounds and not more than 7-1/2 inch-pounds when tested as specified in 4.4.2.1.

3.4.2 Spring. - The spring case, Drawing B81-1-444 which is listed on Drawing C81-1-403, shall rotate freely without binding on the ignition head body, Drawing D81-1-440 which is listed on Drawing C81-1-403, before the nozzle shield is assembled on the gun, and tested as specified in 4.4.2.2.

3.4.3 Trigger. - The trigger, Drawing E81-1-466 which is listed on Drawing C81-1-403, shall move freely and without binding when tested as specified in 4.4.2.3.

3.4.4. Trigger-rod. - When the trigger is pulled back through its full stroke, the end of the trigger-rod, Drawing A81-1-458 which is listed on Drawing C81-1-403, shall not extend less than 1/16 inch beyond the stop lug on the ignition-head body, Drawing D81-1-440 which is listed on Drawing C81-1-403, when tested as specified in 4.4.2.4.

3.4.5 Ignition shield. - The ignition shield assembly, Drawing B81-1-451 which is listed on Drawing C81-1-403, shall screw freely into and out of the ignition head body, Drawing D81-1-440 which is listed on Drawing C81-1-403, when tested as specified in 4.4.2.5. As the shield is screwed the latch, Drawing E81-1-467 which is listed on Drawing C81-1-403, shall engage in the ratchet notches in the ignition shield and prevent return motion. The latch shall finally engage in the locking groove in the ignition shield when the ignition shield is screwed on hand tight.

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3.4.6 Operation. - The ignition head assembly shall be capable of depressing a plunger on the test ignition cylinder (see 6.3) and rotating to the next plunger through five positions when tested as specified in 4.4.2.6.

3.5 Leakage. - The assembled gun shall be tested for leakage as specified in 4.5.

3.6 Workmanship. - All work shall be done in a neat and workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. - A lot shall consist of the guns assembled by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector.

4.2 Sampling for nondestructive inspection. - All guns shall be inspected as specified in 4.3 and tested as specified in 4.4 (see 6.2).

4.3 Inspection.

4.3.1 Components. - The inspector shall ascertain that all materials, components, or subassemblies of the gun covered by separate specifications, standards, or drawings are in accordance with the applicable specifications, standards, or drawings prior to final acceptance of the gun.

4.3.2 Gun. - The inspector shall inspect the gun for compliance with this specification.

4.3.3 Packaging, packing, and marking. - The inspector shall inspect the packaging, packing, and marking of the gun for compliance with section 5.

4.4 Tests.

4.4.1 Valve assembly. - The complete valve assembly shall be inspected for compliance with the requirements of Drawing D81-1-407 which is listed on Drawing C81-1-403 and for the proper adjustment of the valve spring compression which shal be tested as follows:

4.4.1.1 Needle travel. - The operating lever shall be depressed through its full stroke and the position of the point of the needle checked, which shall be not more than 0.040 inch in front of and not more than 0.085 inch in back of the plane of the 5/16-inch diameter hole of the nozzle.

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4.4.1.2 Operating lever. - The force required to move the operating lever through its full stroke shall be determined as follows: Stand the gun in a vertical position with the nozzle uppermost. Release the safety and slowly apply a hook and weight of 40 pounds on the lowest finger notch (farthest from pivot). The valve shall open. In the same manner apply a hook and weight of 15 pounds. The valve shall not open.

4.4.1.3 Safety. - Prior to releasing the safety lever, the operating lever shall remain stationary and the valve closed. When the safety lever is depressed the operating lever will function properly, allowing the valve to open freely.

4.4.2 Ignition head assembly. - The ignition head assembly shall be inspected for compliance with the requirements of Drawing D81-1-1406 which is listed on Drawing C81-1-403 and tested as follows:

4.4.2.1 Spring case. - All spring cases shall be placed on a test fixture which is equipped to hold one case fixed and rotate the other case to wind the spring. The winding shall be done by means of a crank with a pointer and graduated dial for measuring accurately the number of revolutions necessary to wind the spring. The fixture shall be equipped with a balance arm with a sliding counterweight for measuring the torque required to wind the spring (figure 1). The spring shall be tested as follows:

a. Place the spring in the fixture and wind to determine that it can be wound, under tension, at least 2-1/4 turns.

b. Release the tension on the spring by turning the crank until the spring case is free from tension.

c. Set the counterweight at the inner (2-1/2 inch-pound) position.

d. Turn the crank to wind the spring. As soon as all lost motion is taken out and spring tension can be felt on the shaft, note the position of the crank pointer. Any motion of the crank beyond this point shall cause the balance arm to lift, indicating a spring torque of greater than 2-1/2 inch-pounds.

e. Slide the counterweight to the outer (7-1/2 inch-pound) position.

f. Continue turning the crank until it has been turned two full revolutions from the position noted above (d). The balance arm shall not lift, indicating a spring torque of less than 7-1/2 inch-pounds.

g. Relieve the tension in the spring by turning the crank before removing the spring case from the test fixture.

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4.4.2.2 Spring. - The freedom of movement of the spring case on the ignition-head body shall be tested as follows: Remove the ignition shield and rotate the spring case to see that it turns freely on the ignition-head body.

4.4.2.3 Trigger. - The force required to move the trigger through its full stroke shall be determined as follows: Hold the gun in a vertical position with the nozzle end up. Suspend a hook and weight of 6 pounds on the center of the trigger finger notch. The trigger shall be moved through its full stroke with the weight.

4.4.2.4 Trigger-rod. - The maximum travel of the trigger-rod shall be determined as follows: Pull the trigger back through its full stroke and measure the distance from the end of the trigger-rod to the stop lug on the ignition head body.

4.4.2.5 Ignition shield. - The fit of the ignition shield and operation of the latch shall be tested as follows: Screw the ignition shield on the ignition head body. As the ignition shield is turned, see that it turns freely and without binding, that the latch engages in the ratchet notches and prevents return motion, and that the latch engages in the locking groove when the ignition shield is screwed on hand tight.

4.4.2.6 Operation. - The operation of the ignition-head assembly shall be tested as follows: Place a test ignition cylinder (see 6.4) over the end of the barrel. Turn the ignition cylinder and the spring case in a clockwise direction as far as they will turn freely. Place the ignition shield in position and screw it on to the ignition head. The threads on the ignition shield must be engaged with the threads of the ignition head during the first revolution of the ignition shield. It may be necessary, in some cases to turn the ignition shield backwards a short distance to engage the threads before screwing the shield all the way on. During the last revolution of the ignition shield the latch shall engage at least 2 of the ratchet notches on the ignition shield to prevent it from unwinding. When the ignition shield is screwed on hand tight the latch shall drop into the locking groove in the ignition shield. Pull back the trigger. At each stroke of the trigger a test cylinder plunger shall be pushed in and the test cylinder shall index one position. The spring shall have sufficient stored energy to index the test cylinder through five positions. This operation of winding the ignition shield on and pulling back the trigger five times shall be performed three times.

4.5 Leakage of assembled gun. - The assembled gun, less handle, shall be subjected to an hydrostatic pressure of 750 p.s.i. using diesel fuel conforming to Specification VV-D-361 and observed for leakage.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 For shipment to assembly plants.- The flame thrower gun shall be packaged as specified on Drawing B81-12-20 except that the desiccant and moisture-vaporproof barrier will not be required. The package shall be closed with gummed paper tape conforming to Grade B of Specification UU-T-111 to facilitate reuse of the box.

5.1.2 For shipment as spare parts.- The flame thrower gun shall be packaged as specified on Drawing B81-2-20. The sealed package shall then be packaged in a W6s, style RSC fiberboard box conforming to Specification JAN-P-108.

5.2 Packing.

5.2.1 For shipment to assembly plants.- The flame thrower gun, packaged as specified in 5.1.1, shall be packed as specified in Specification MIL-P-10577.

5.2.2 For shipment as spare parts.- Three flame thrower guns, packaged as specified in 5.1.2, shall be packed in a style 2, 2-1/2, or 4 nailed wood box, conforming to Specification JAN-P-106.

5.3 Marking.- In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type of packing required (see 5.2).

6.2 Sampling and testing.- When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify in whole or in part the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.3 Test ignition cylinder.- The test ignition cylinder will be furnished by the contracting officer for use in making the operation tests of the ignition-head assembly.

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Custodian:

Army - Chemical Corps

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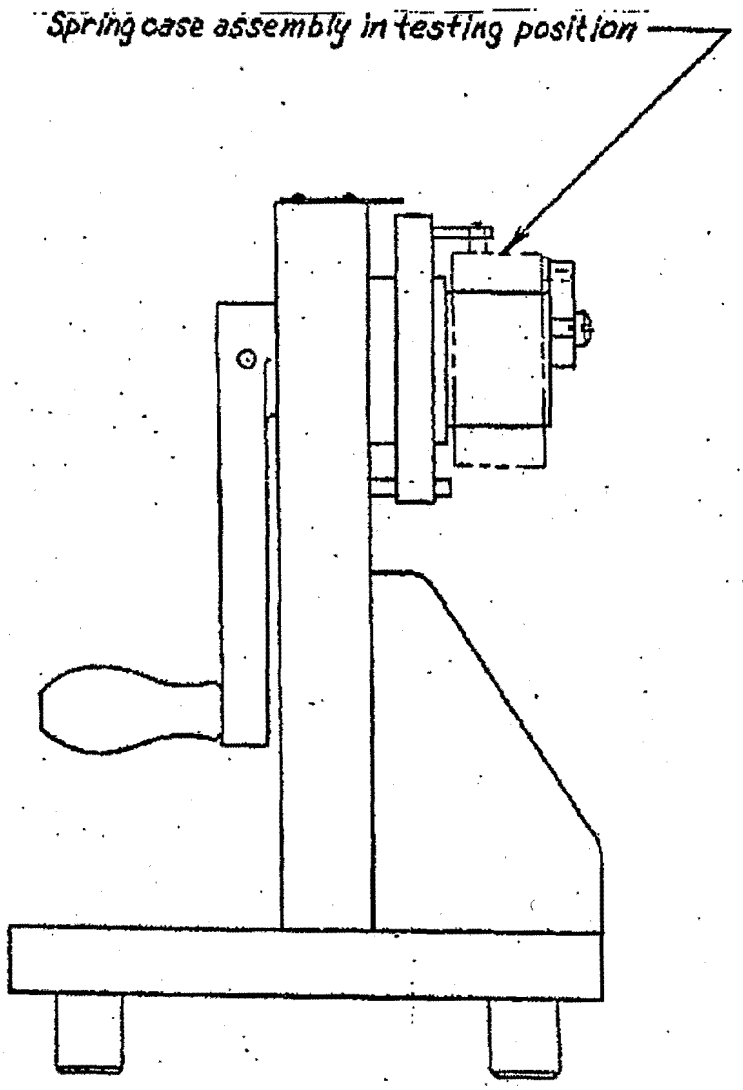


Figure 1. – Spring torque testing apparatus.

For assembly see CNL C Drawing C24-116-15

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